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Paleomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and the opening of the Iapetus Ocean

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Abstract

The Skinner Cove Formation of western Newfoundland is an alkali volcanic suite transported in the Ordovician onto Laurentia's Iapetus Ocean margin. It has previously yielded a U–Pb zircon age of 550.5^{+3}_{-2} Ma, has no penetrative deformation and has only been affected by zeolite facies metamorphism. At 10 sites, flows and dykes of the Skinner Cove Formation exhibit a stable characteristic 'A' remanence carried by magnetite. This 'A' remanence is shown to be primary by an intraformational conglomerate test using 20 trachybasalt clasts (whereas 11 alkali basalt clasts were overprinted, probably because they originally contained titanomagnetite from which rutile exsolved soon after burial by overlying flows). After tilt-correction, the mean 'A' remanence direction is southeast and down ($D = 144^{\circ}$, $I = 32^{\circ}$; $\alpha_{95} = 11^{\circ}$, $k = 21$, $N = 10$ sites). The paleolatitude calculated from the ten 'A' site virtual geomagnetic poles is $19^{\circ}S \pm 9^{\circ}$, which likely represents the paleolatitude of Laurentia's Iapetan margin at ~550 Ma. Evidence that the Skinner Cove Formation originated at this margin includes its structural position within the Humber Arm Allochthon, its within-plate trace element geochemistry, and its similarity in age to other alkali magmatism that more certainly represents Laurentia's Iapetan margin. Comparison with other Laurentian paleomagnetic data implies that Laurentia drifted very rapidly northward from the south polar region to the equator between ~570 and 550 Ma. We suggest that the start of this rapid northward drift at ~570 Ma marked the onset of Iapetus sea-floor spreading between Laurentia and West Gondwana. Further well-constrained paleomagnetic data of ~570 to ~550 Ma age are needed from Laurentia to test its high rate of northward drift, and from West Gondwanan cratons to test that they remained relatively stationary. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; paleolatitude; upper Proterozoic; Laurentia; Iapetus; plate tectonics

1. Introduction

Breakup and reorganization of supercontinental assemblies characterized the late Neoproterozoic [1] and may have influenced the early evolution of macroscopic animals [2]. The opening of the Iapetus Ocean, creating an east–west subtropical seaway, may have been a key paleogeographic change which aided the diversification of life [3].

Recently, several late Neoproterozoic continental reconstructions have been proposed [2,4,5], offering insights on worldwide paleogeographic and paleoclimatic conditions during this seminal period of Earth's history. However, additional Neoproterozoic geological and paleomagnetic data are needed to better constrain the relative positions and movements of

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cratons involved in the breakup of the supercontinent Rodinia [6], and in the final dispersal of the proposed late Neoproterozoic supercontinent Pannotia [2]. The need is especially acute for a craton such as Laurentia, which likely had a high absolute plate velocity during the dispersal [7,8].

A constraint on the terminal Neoproterozoic position of Laurentia was suggested by preliminary paleomagnetic data from the 550 Ma Skinner Cove Formation in western Newfoundland [9,10]. We now report in full on the paleomagnetism, showing that it very likely provides a paleolatitude estimate for Laurentia's Iapetan margin at the end of the Neoproterozoic. We suggest that Laurentia straddled the equator by ~550 Ma, having moved there from high southerly latitudes at ~570 Ma during a period of rapid northward drift which opened the Iapetus Ocean between Laurentia and Gondwana.

2. Geological setting

In northeastern North America, rocks of the Humber Zone and failed rift basins (Fig. 1) record the history of the ancient Laurentian continental margin. Much of this history is recorded in western Newfoundland, from late Neoproterozoic Iapetus rifting to mid-Ordovician foundering of its carbonate platform during obduction of Taconic allochthons. The largest of these is the Humber Arm Allochthon, an imbricate stack of transported slices of Laurentian shelf slope sediments overlain by the Skinner Cove volcanics slice, which is in turn overlain by late Cambrian–early Ordovician igneous slices of island arc and back-arc affinity [11].

The Skinner Cove Formation (Fig. 2) is an ~800 m thickness of well-bedded volcaniclastic rocks and lava flows dipping ~70° to the southeast, fed by subvolcanic dykes (that become subvertical when tilt-corrected). The volcanics are a differentiated alkalic suite of ankaramite, alkali basalt, trachybasalt and trachyte in three members [12]. Most of the Skinner Cove Formation is remarkably unaltered, with no penetrative deformation and evidence of only zeolite facies metamorphism [12].

The Skinner Cove Formation has been interpreted to represent oceanic islands or seamounts near Laurentia [12] or rift-facies volcanics related to the

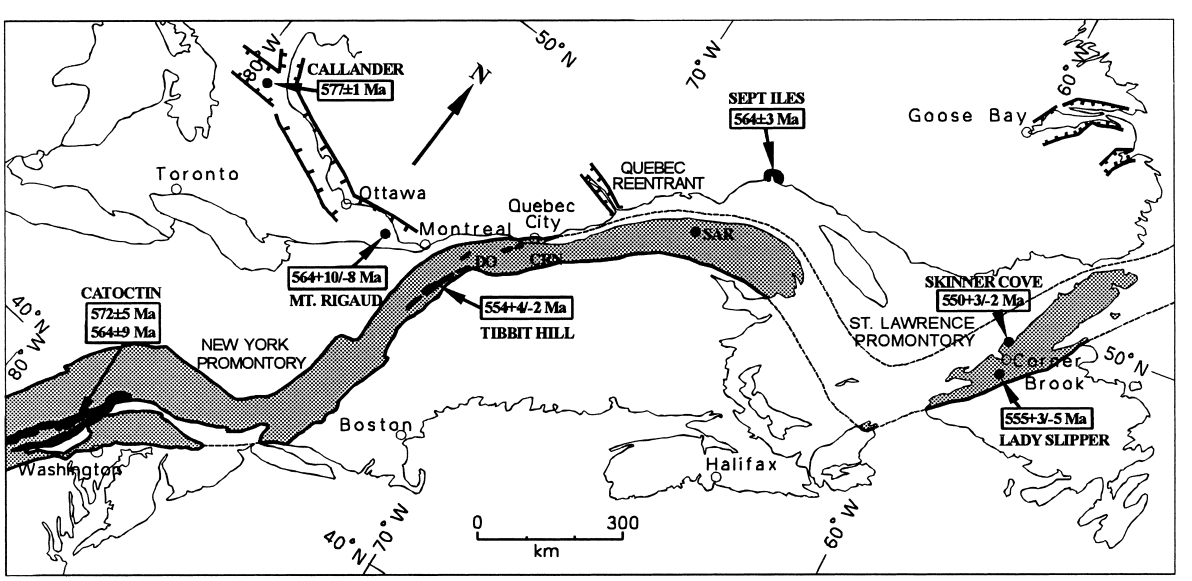


Fig. 1. Latest Neoproterozoic magmatism and rift-related features of the Appalachians. The Humber Zone (shaded) represents the Iapetus Ocean margin of Laurentia onto which the Skinner Cove Formation was transported in the mid-Ordovician. All ages shown are U–Pb zircon dates with 2σ errors (references in text). Abbreviations for Quebec Humber Zone alkali volcanics are: *DO* = Drummondville Olistostrome; *CRN* = Chaudierre River Nappe; *SAR* = St. Anne River Nappe.

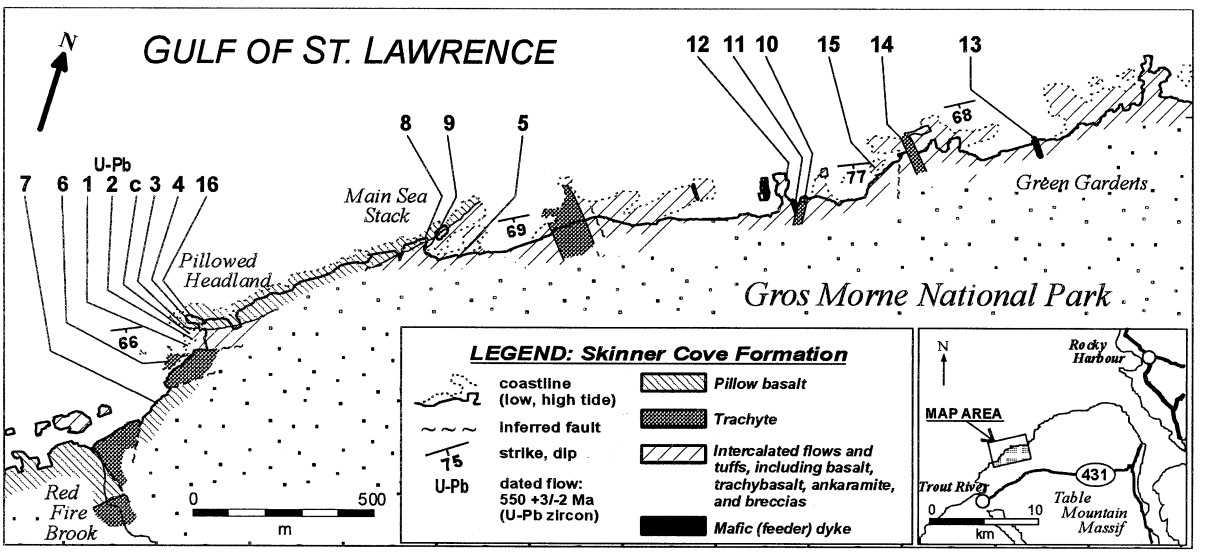


Fig. 2. Sites where the Skinner Cove Formation was sampled in a structural slice of the Humber Arm Allochthon of western Newfoundland. Geology after Baker [12]. See Fig. 1 for regional context.

opening of the Iapetus Ocean [11,12]. The age of the Skinner Cove Formation is 550.5^{+3}_{-2} Ma based on U–Pb dating of zircons from an ankaramite flow at site 2 [10].

3. Paleomagnetic methods and results

All paleomagnetic sites are in the Main Sea Stack Member, except sites 6 and 7 which are in Red Fire Brook Member trachyte and Wallace Brook Member pillow basalts respectively. At least six oriented block samples were collected from each of the 16 sites. Also, 36 oriented block samples were collected for an intraformational conglomerate test at site C. Samples were oriented with the aid of a sun compass. Strike and dip at each site were determined from interbedded tuffs. Cylindrical specimens (2.4 cm diameter, 2.1 cm length) were cut from the oriented block samples in the laboratory.

All oriented specimens used in the study were stepwise demagnetized. Remanence was measured between steps using a Schonstedt SSM-1 spinner magnetometer. Stepwise alternating-field (AF) demagnetization using a Schonstedt GSD-1 demagnetizer was applied to one specimen from each block sample. Stepwise thermal demagnetization using a Schonstedt TSD-1 demagnetizer was applied to at least one specimen per site.

Tilt-corrected vector plots of the stepwise demagnetizations (Fig. 3) underwent principal-component analysis [13]. A trachybasalt specimen from site 3 (Fig. 3a) exhibits an initial change in direction during AF demagnetization with the removal of a small component that is likely a viscous remanent magnetization (VRM). Above ~15 mT, the remanence maintains a southerly down direction as intensity decreases towards the vector plot origin. Thermal demagnetization of a second specimen from the same block sample shows similar behaviour (Fig. 3b), with remanence above ~250°C showing little change in its southeasterly down direction. A pair of alkali basalt specimens from a site 4 block sample show broadly similar demagnetization characteristics, with a southeast and down magnetization isolated at coercivities higher than ~15 mT (Fig. 3c) and at unblocking temperatures higher than ~250°C (Fig. 3d).

A VRM is resolvable in 47 of our 96 block samples. It has a north, steep down in-situ direction (mean $D = 351°$, $I = 72°$, $\alpha_{95} = 8°$, $k = 36$, $N = 10$ sites), similar to the present Earth's field at Skinner Cove ($D = 335$, $I = 72°$) and likely of recent origin. VRM usually appears to be fully

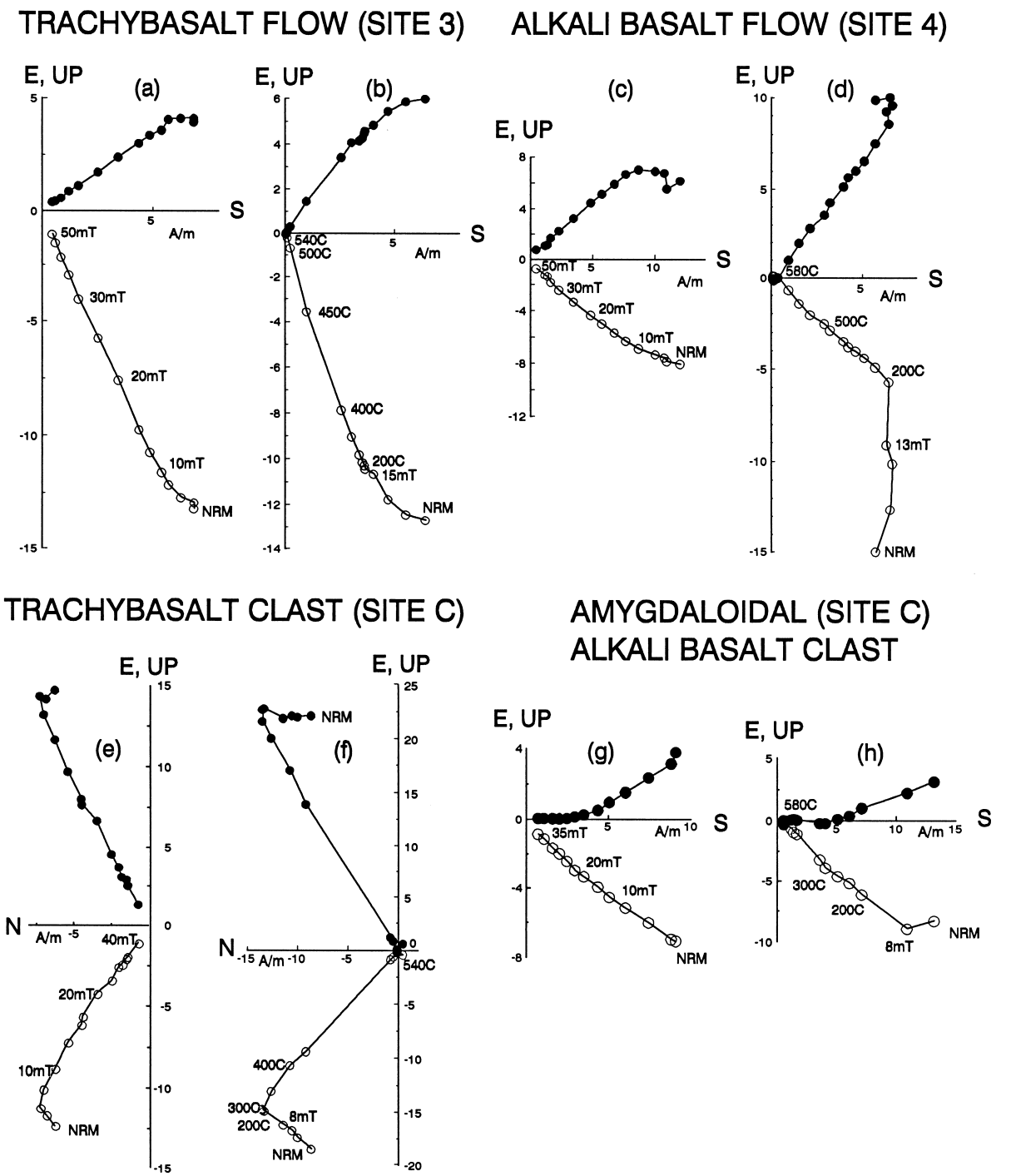


Fig. 3. Comparison of alternating field (*AF*) demagnetizations with thermal demagnetizations for specimens from the same block sample for site 3 trachybasalt, site 4 alkali basalt and for a trachybasalt clast and an amygdaloidal alkali basalt clast from an intraformational conglomerate (site C). Thermal demagnetizations were preceded by partial AF demagnetization. The tilt-corrected orthogonal vector plots show decay projected in the UP-S (E–S) plane with open (closed) circles.

Table 1
Skinner Cove formation paleomagnetic results

	Site	Rock type	Str (º)	Dip (º)	*N*	A/T	*D* (º)	*I* (º)	*D*′ (º)	*I*′ (º)	α_{95} (º)	*k*
A:	6	Trachyte	073	80	6	A&T	17.2	78.2	155.9	19.7	14.0	23.7
	1	Ankaramite	057	70	5	A	98.3	89.3	146.5	19.5	12.0	41.9
	3	Trachybasalt	062	67	6	A&T	341.3	56.3	142.7	56.0	9.2	54.4
	4	Alkali basalt	062	67	5	A	7.4	71.4	138.5	37.6	5.6	188.9
	9	Alkali basalt	044	69	6	A	314.8	72.7	133.6	38.3	10.7	40.4
	5	Alkali basalt	044	69	6	A&T	310.6	51.0	138.3	59.8	13.9	24.1
	12	Ankaramite	053	62	2 [a]	A&T	205.4	81.8	150.9	24.0		
	11	Mafic dyke	057	56	4	A&T	207.6	80.3	156.7	28.8	17.5	28.6
	10	Trachyte dyke	057	56	3 [a]	A&T	151.2	70.1	148.8	14.1	30.2	17.8
	15	Trachybasalt	059	77	5	A&T	345.0	79.4	145.8	23.2	8.5	81.9
	14	Trachyte dyke	059	72	9	A&T	219.1	87.8	151.1	17.2	11.1	22.4
	13	Mafic dyke	050	80	5	A&T	16.1	77.2	128.9	16.9	14.5	28.7
Mean of 10 ‘A’ sites, in situ							336.7	78.0			9.7	25.9
Mean of 10 ‘A’ sites, tilt corrected									144.4	31.5	10.8	21.1
B:	7	Alkali basalt	020	78	4	A	143.2	−38.2	247.3	−50.5	12.7	53.0
C:	8	Alkali basalt	044	69	5	A&T	153.5	18.4	161.2	−46.2	20.7	14.6
	16	Mafic dyke	062	67	1	T	151.8	23.6	151.8	−43.4		

[a] Sites 10, 12 not used for calculating mean of ‘A’ sites due to insufficient number of useable samples.
Str, Dip: strike, dip (in degrees) of bedding; *N*: number of samples contributing to site mean; A/T: AF or thermal demagnetization; *D*, *I*: declination and inclination in situ; *D*′, *I*′: declination and inclination after tilt correction; α_{95}: circle of 95% confidence about mean direction; *k*: precision parameter.

removed by AF demagnetization to 20 mT or thermal demagnetization to 300ºC.

For 25 samples, the existence of any component beyond a VRM could not be resolved due to remanence above 25 mT or 350ºC being unstable or too weak to be measured. In the other 71 samples, a remanence of stable direction is resolvable by least squares line fitting to vector plots, excluding coercivities below 20 mT and unblocking temperatures below 300ºC to avoid VRM. A tilt-corrected southeast and down characteristic remanence (to be called ‘A’) is resolved in 62 of the stable samples, of which 57 contribute to 10 ‘A’ site means (Table 1). The ‘A’ remanence usually decays linearly towards the origin of a vector plot and is fully removed between 520ºC and 580ºC, showing that it is carried by magnetite.

Stable remanence components carried by magnetite with directions other than ‘A’ have been identified in only ten samples (Table 1). The hard ‘B’ remanence, found in four samples at site 7, has demagnetization characteristics similar to the ‘A’ remanence but its west southwest and up tilt-corrected direction is anomalous. The hard ‘C’ remanence, found in the five samples from site 8 and one sample from site 16, is poorly grouped in a south–southeast, shallow down in-situ direction that may represent a Paleozoic post-folding overprint.

4. Evidence for primary remanence

A conglomerate provided the only conclusive test for primary remanence. A fold test proved inconclusive, as expected since there is little variation in bedding attitude, resulting in no significant change in clustering of the ten ‘A’ site mean directions upon untilting. Although Beaubouef et al. [9] argued that the characteristic remanence is probably of pre-folding age, this involved comparing the Skinner Cove volcanics with dubious correlatives [12].

An intraformational conglomerate test was performed using 21 trachybasalt clasts and 15 amygdaloidal alkali basalt clasts in a reworked tuff (site C). The trachybasalt clasts closely resemble and may have been drawn from the stably magnetized site 3 trachybasalt flow which lies ∼2 m stratigraphically below the clasts. Amygdaloidal alkali basalt clasts were much more amygdaloidal than samples from

the stably magnetized site 4 flow (which lies ~15 m stratigraphically below the conglomerate) or any other alkali basalt sites. The tuff is overlain by two ankaramite flows (site 2 and 1) which lie stratigraphically ~10 m and ~15 m respectively above the sampled clasts, with site 2 yielding the U–Pb date but only site 1 yielding a useable remanence.

Trachybasalt clast specimens (Fig. 3e,f) typically display two well-resolved components of magnetization. A soft component is removed by ~15 mT (or ~250ºC). It is resolved in 13 clasts and has a northwest and steep down in-situ direction ($D = 310^{o}$, $I = 81^{o}$, $\alpha_{95} = 22.5^{o}$, $k = 4.3$; $N = 13$), that is close to the present Earth's field direction and is likely a recent VRM. A hard component resolved in 20 clasts has coercivities mainly greater than 30 mT and shows a large scatter of directions (Fig. 4a). The 20 hard component directions pass a statistical test for randomness [14]. (The specimens' unit vector sum $R = 5.38$ is less than the $R_0 = 7.17$ expected for 20 specimens at the 95% confidence level). Within-clast consistency is shown by the three clasts for which a second specimen was thermally demagnetized and found to have the same direction (above ~250ºC) as the hard component of its neighbour.

Amygdaloidal alkali basalt clast specimens (Fig. 3g,h) carry a soft component that is removed by demagnetization to ~20 mT (or ~300ºC), and is likely a recent VRM, having a mean in-situ direction ($D = 350^{o}$, $I = 83^{o}$, $\alpha_{95} = 4.8^{o}$, $k = 65$, $N = 15$) resembling the present Earth's field direction. A hard component of magnetization is also resolvable in 11 of the 15 amygdaloidal clast specimens. The hard component directions scatter loosely (Fig. 4b), but are not random ($R = 9.33$ is greater than $R_0 = 5.28$ expected for 11 specimens at the 95% confidence level). The hard remanence (with tilt-corrected mean $D = 145$, $I = 6^{o}$; $\alpha_{95} = 20^{o}$, $k = 6$, $N = 11$) is thus mostly an overprint. Within-clast consistency is shown by the three clasts for which a second specimen was thermally demagnetized and shown to have the same direction (above ~300ºC) as the hard component of its neighbour.

The conglomerate test (Fig. 4a) proves that magnetite-borne remanence in trachybasalt with coercivities above 15 mT and unblocking temperatures above ~300ºC was acquired before incorporation of the trachybasalt clasts into the intraformational tuff. Hence the 'A' remanence direction found in the underlying site 3 trachybasalt flow and other 'A' sites dates from the 550 Ma age of the volcanics and is likely a primary thermal remanence.

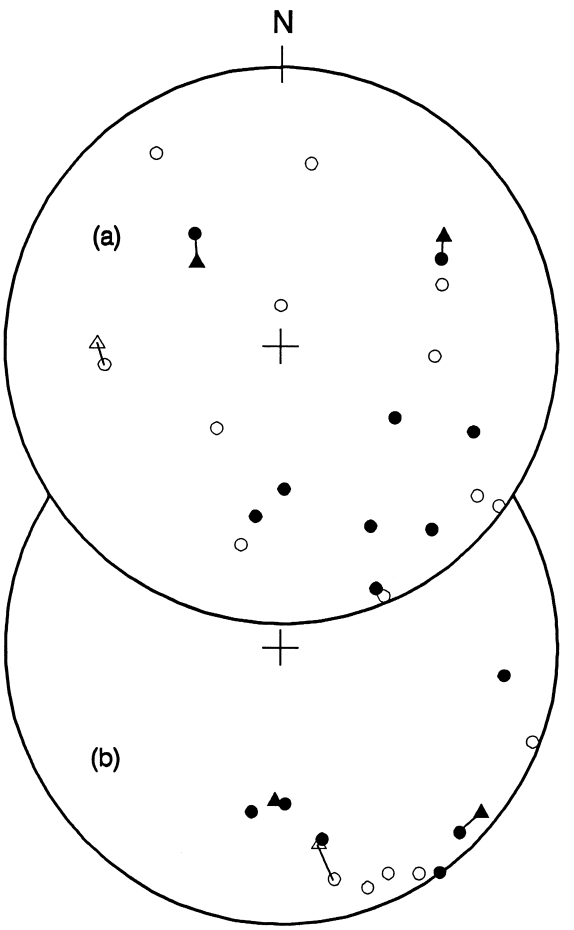


Fig. 4. Conglomerate tests at site C using hard component directions from (a) 20 trachybasalt clasts, and (b) 11 amygdaloidal alkali basalt clasts. In their tilt-corrected equal area stereoplots, down (up) directions are marked by closed (open) circles. Triangles (joined to circles) indicate a second specimen from the same clast that was thermally rather than AF demagnetized. The trachybasalt clasts carry a primary remanence. The amygdaloidal alkali basalt clasts were remagnetized, probably soon after burial by overlying flows.

The amygdaloidal alkali basalt clasts carry an overprint that is likely thermo-chemical rather than thermoviscous since neighbouring trachybasalt clasts with similar high coercivity and high blocking tem-

perature magnetite have not been thermoviscously overprinted. The overprint direction ($D = 145^{o}$, $I = 6^{o}$) is similar to the 'A' remanence of the overlying site 1 ankaramite flow ($D = 147$, $I = 20$) and was likely acquired at ~550 Ma due to heating by the overlying flows. The zeolite facies metamorphism implies that this heating was to 200ºC or less [12].

We suggest the following thermochemical mechanism for overprinting the amygdaloidal basalt clasts without affecting the trachybasalt clasts. Although both basalts crystallized with titanomagnetite of high titanium content, the trachybasalt titanomagnetite underwent the usual oxidation-exsolution during initial cooling, resulting in a fine intergrowth of ilmenite and magnetite with Curie point near 580ºC [15] and acquisition of a thermal remanence of high blocking temperature. In contrast, the titanomagnetite in the amygdaloidal basalt remained mostly unexsolved with a Curie point near 150ºC as is sometimes observed in submarine basalts [16]. The amygdaloidal basalt clasts were incorporated into the conglomerate with too low a Curie point for stable remanence. Heating by overlying flows soon caused zeolite facies metamorphism that induced titanomagnetite to exsolve ferri-rutile granules, raising the Curie point to ~580ºC (as previously observed [15]) and stably magnetizing the clasts in a direction similar to the overlying flows.

Support for this suggestion comes from comparing polished sections of one amygdaloidal basalt clast and one trachybasalt clast used in the conglomerate test. They were observed with a scanning electron microscope capable of semi-quantitative analysis. The amygdaloidal basalt contains large grains of magnetite with ferri-rutile granules in the cores and rutile grains around the margin (Fig. 5). In contrast, magnetite in the trachybasalt clast contains ilmenite lamellae, but no rutile granules were observed. Rutile granules were also absent or uncommon in magnetite in the four other polished sections examined from sites 3, 4, 10, and 13, representing the four rock types carrying 'A' remanence.

In conclusion, the hard 'A' remanence recognized in flows and dykes of the Skinner Cove Formation is primary and likely a thermal remanence. The mean characteristic direction for each of the ten 'A' sites as listed in Table 1. Their mean tilt-corrected direction

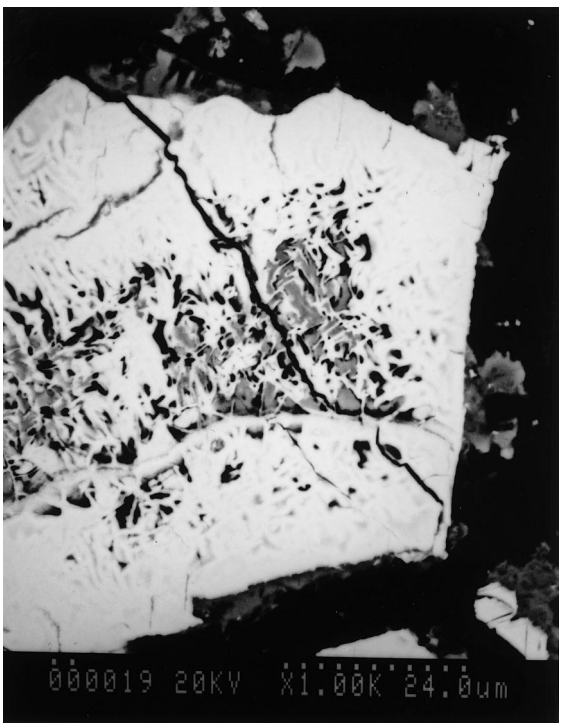


Fig. 5. Scanning electron microscope backscatter image from an amygdaloidal basalt clast used in the conglomerate test of Fig. 4b. The dotted scale bar represents 24 μm. Silicate minerals (black) enclose a large grain of magnetite (white) with rutile (grey) in granules in its core and in grains around its margin. The rutile likely exsolved from original titanomagnetite during zeolite facies metamorphism, raising the Curie point and stably magnetizing the clast soon after burial by overlying flows.

(Fig. 6) is $D = 144.4^{o}$, $I = 31.5^{o}$ ($\alpha_{95} = 10.8^{o}$, $k = 21.1$, $N = 10$ sites). Although the in-situ mean direction ($D = 337^{o}$, $I = 78^{o}$) is similar to the present Earth's field direction at Skinner Cove ($D = 335^{o}$, $I = 72^{o}$), the hard 'A' remanence cannot be a recent (or Paleozoic) remagnetization because of the positive conglomerate test for trachybasalt clasts, as discussed above. Calculation of a mean pole and A_{95} from the ten 'A' site virtual geomagnetic poles gives a paleolatitude of 19ºS ± 9º. (Beaubouef et al. [9] reported ~12ºS). Note that the corresponding paleopole, quoted in Table 2, must be used with caution since we cannot correct for any rotation about an axis perpendicular to the bedding plane undergone by the Skinner Cove slice.

Table 2
Paleolatitude of Laurentia's Iapetan margin between 580 Ma and 510 Ma [a]

Unit	Age (Ma)	Paleopole [b]	A_{95} (°)	K	Paleolatitude (°S)
Moore Hollow Gp., TX	505–523 bio	1°S 163°E	5	94	31 ± 5
Tapeats SS, TX	~525 bio	5°N 158°E	3		27 ± 3
Skinner Cove Fm, Nfld	550^{+3}_{-2} U–Pb	(15°N 157°E) [c]	9	31	19 ± 9
Johnnie Rainstorm, NV	~555?	10°S 162°E	~10	–	−38 ± 10
Sept Iles 'B', Quebec	564 ± 3 U–Pb	44°S 135°E	10	16	80 ± 10
Catoctin Basalts 'A', VA	564 ± 9 U–Pb	43°S 118°E	17	17	86 ± 17
Callander Complex, ONT	577 ± 1 U–Pb	46°S 121°E	6	25	87 ± 6

[a] Paleolatitude calculated for present Skinner Cove location (49.5°N, 302°E) using published poles for Laurentia (see text for references).
[b] North paleopole calculated as the mean of site virtual geomagnetic poles, with A_{95} being the circle of 95% confidence about the mean pole direction, and K being the precision parameter.
[c] Skinner Cove Fm is assumed to represent Laurentia, but any rotation about an axis perpendicular to its bedding plane cannot be corrected for.
Ages obtained from biostratigraphic (bio) or radiometric (U–Pb) methods.

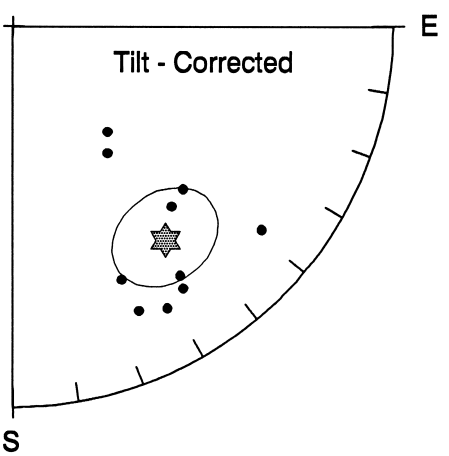


Fig. 6. Equal area stereoplot of the ten tilt-corrected 'A' component site mean directions (Table 2). Down directions are marked by closed circles. The star shows the mean direction ($D = 144°$, $I = 31.5°$; $\alpha_{95} = 10.8°$, $k = 21.1$, $N = 10$ sites) with its circle of 95% confidence.

5. Evidence for Laurentian affinity

Because the Skinner Cove Formation is allochthonous and unfossiliferous, we can show only that it likely (but not certainly) originated near the Iapetan margin of Laurentia, providing a paleolatitude estimate for this margin at 550 Ma.

Within the Humber Arm Allochthon imbricate stack of structural slices, the Skinner Cove Formation slice overlies slices of Laurentian shelf slope sediments [11], and is overlain by igneous slices. The topmost igneous slice is the extensive 486 Ma Bay of Islands ophiolite, which correlates with ophiolites of very similar age and geochemistry in the adjoining Notre Dame Subzone of central Newfoundland [17] that are conformably overlain by sediments containing Arenig fossils of Laurentian affinity [18]). An imbricate order of assembly [11] places the Skinner Cove volcanics slice inboard of the Bay of Islands back-arc ophiolite and other igneous slices prior to their mid-Ordovician obduction onto Laurentia as part of the Humber Arm Allochthon. The structural position of the Skinner Cove Formation slice, its complete lack of penetrative deformation [12] despite being thin (~0.8 km) and its low grade of metamorphism suggest that the Skinner Cove Formation has not been transported far, having instead formed close to Laurentia's Iapetan margin.

The Skinner Cove volcanics are strongly alkalic, with the basalts having high TiO_2 content (averaging 3.2% by weight), high Zr/Y and Nb/Y ratios and elevated light rare earth element (LREE) abundances relative to chondrite [12]. Tectonic discrimination plots (e.g. [19]) using immobile elements Zr, Y, Nb (Fig. 7) suggest the Skinner Cove Formation basalts formed in a within-plate setting, either associated with continental rifting, or with a mantle plume. Subaqueous eruption of some units, locally steep paleoslopes and lack of continentally-derived sediments led Baker [12] to favour an oceanic volcanic setting for the Skinner Cove Formation.

Titaniferous alkali basalts with similarly high

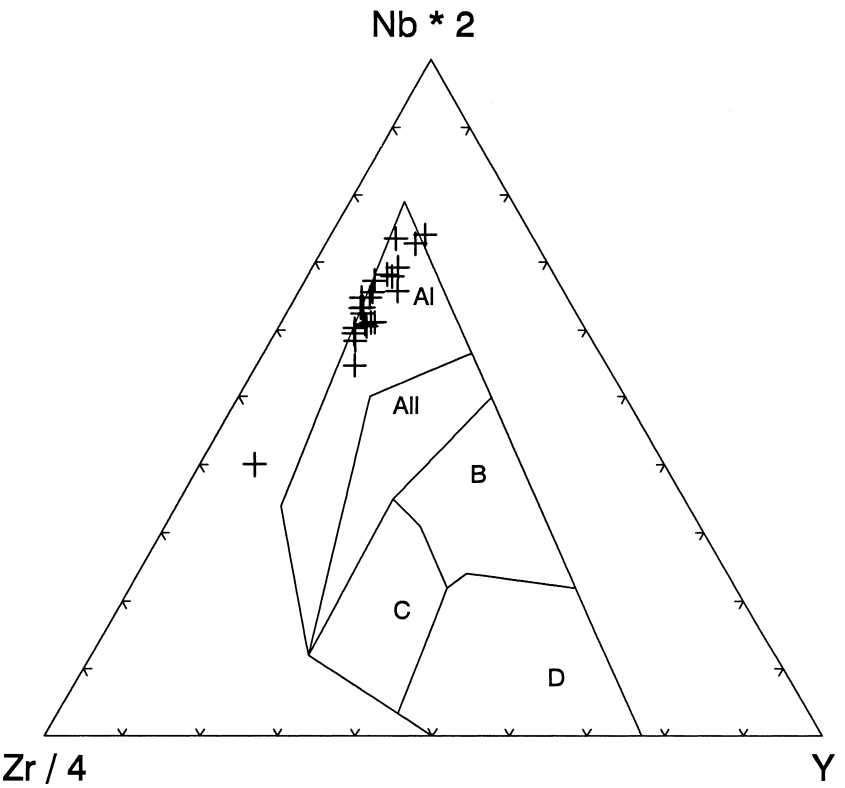


Fig. 7. Tectonic discrimination plot [19] using trace elements Zr, Y and Ti which indicates a within-plate alkali tectonic setting for the 23 alkali basalt samples from the Skinner Cove Formation analysed by Baker [12]. *Al*, *All* = within plate alkali basalt; *All*, *C* = within plate tholeiite; *B*, *D* = mid-ocean ridge basalt; *C*, *D* = volcanic arc basalt.

Zr/Y, Nb/Y and enriched LREE are found intercalated with Laurentian margin sediments in allochthonous units along the Quebec Humber Zone (e.g., the Chaudiere River Nappe and Drummondville Olistostrome [20], and the Ste-Anne River Nappe [21]). These alkalic units are interpreted to be lateral equivalents of the Tibbit Hill metavolcanics [22,23], formed originally as Laurentia-related seamounts [23] similar to the Skinner Cove volcanics (Fig. 1).

The Tibbit Hill metavolcanics yield a 554^{+4}_{-2} Ma U–Pb zircon date [24], and have a confirmed continental setting, resting on Grenville basement [22] near the junction of the Iapetan margin with the Ottawa–Bonnechere failed rift arm [23]. In the Humber Zone of western Newfoundland, tonalitic gneiss of the Lady Slipper Pluton was dated at 555^{+3}_{-5} U–Pb (U–Pb zircon), contains a ~1500 Ma inheritance from local Grenville basement, and is disconformably overlain by Laurentian passive margin sediments [25]. The Lady Slipper Pluton may represent crustal melting in response to regional heating from an underlying mantle plume.

Similarities of age, trace element geochemistry and tectonic setting favour a common origin for the Tibbit Hill metavolcanics, Quebec Humber Zone alkali volcanics, and Skinner Cove volcanics as part of a regional magmatic (plume) event along the Laurentian margin.

6. Laurentian Neoproterozoic paleolatitude change

Fig. 8 presents paleolatitudes between 580 Ma and 510 Ma for the Newfoundland segment of the Laurentian margin, calculated from published Laurentian paleopoles (Table 2) and assuming the Skin-

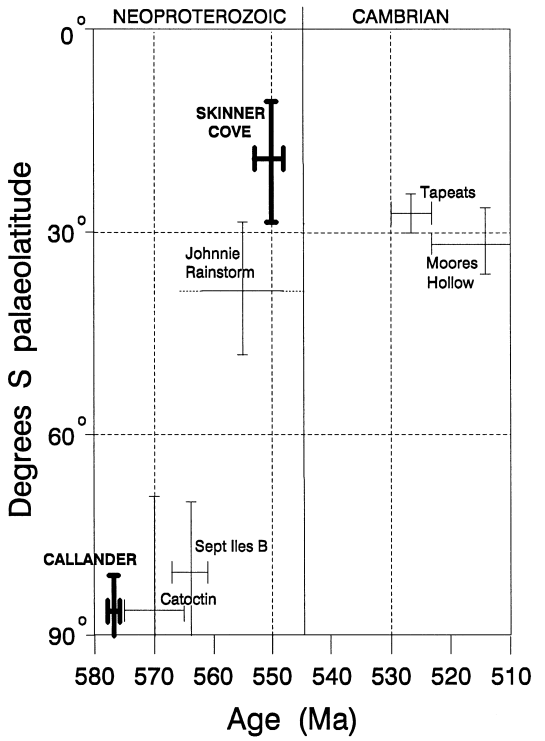


Fig. 8. Paleolatitude plot of the west Newfoundland segment of Laurentia's Iapetan margin calculated from late Neoproterozoic–Cambrian Laurentian paleopoles (Table 2). Bold type indicates that primary remanence has been proven. We assume that the Skinner Cove Formation originated near Laurentia's Iapetan margin (which is very likely, but not certain). We suggest that the start of rapid northward drift of Laurentia at ~570 Ma represents the start of Iapetus sea-floor spreading.

ner Cove Formation represents this margin at 550 Ma.

Recently tabulated Laurentian poles of ca. 550 Ma age [5,26] typically have poor control on the age of magnetization. For example, the Long Range Dykes of Labrador yield a K–Ar (biotite) date of 553 ± 22 Ma, that has been interpreted [26] as the age of a remagnetization, producing a low-inclination remanence. However, a baked contact test suggests this low-inclination remanence corresponds [27] to the 615 ± 2 Ma and 614^{+6}_{-4} Ma U–Pb dates for the dykes [28]. The Double Mer Formation red beds of Labrador [27] have a similar low-inclination remanence but poor age control. Hence, the Long Range Dykes, and Double Mer Formation are excluded from Fig. 8, as are the Buckingham volcanics of Quebec that are now known to be Mesoproterozoic [29].

Sandstones of the dual-polarity Johnnie Formation (Rainstorm Member) of Nevada are estimated to be ~10 million years (Myr) older than the base of the Cambrian (cf. [26]), giving an age of ~555 Ma (using the timescale of Ref. [30]). Their (unrotated) pole [31] implies a low southerly paleolatitude for Laurentia at ~555 Ma (Fig. 9), supporting our ~550 Ma Skinner Cove Formation result. Early to mid Cambrian poles (Table 2) summarized in Ref. [5], imply that Laurentia remained at low paleolatitudes onwards through the Cambrian.

The Callander Complex of Ontario has a U–Pb zircon age of 577 ± 1 Ma [32] and carries a primary remanence as demonstrated by a baked contact test [33]. This places the Iapetan margin of Laurentia near the south pole at 577 Ma. The Catoctin volcanics 'A' remanence was acquired before mid-Ordovician folding [26] and its U–Pb ages (572 ± 5 Ma and 564 ± 9 Ma, [34]) suggest that the margin of Laurentia remained near the south pole until ~570 Ma. The dual-polarity Sept Iles 'B' remanence [35] suggests that Laurentia had begun moving northward by 564 ± 3 Ma [36].

Assuming that Laurentia's northward drift began within 7 Myr of 570 Ma and ended at 550 Ma implies a very high a drift rate of ~34 cm/year (twice that of India between 80 Ma and 50 Ma [37]). This supports previous proposals of rapid northward drift of Laurentia between 580 Ma and 540 Ma [7], at interpolated minimum rates as high as 23 cm/year [8].

Laurentia's very high drift rate implied by Fig. 8 could be halved if the Skinner Cove Formation originated ~30° north of Laurentia's Iapetan margin to which it was moved during Iapetus closure. However, this seems to require the Skinner Cove slice to be well south of Laurentia by the late Cambrian so that oblique closure of the Iapetus could move it at least ~2000 km along Laurentia's southern margin before mid-Ordovician obduction. That the Skinner Cove Formation represents so far-travelled a terrane is unlikely but possible, and deserves future testing.

Another way to avoid the very high drift rate implied by Fig. 8 is to ascribe the rapid paleolatitude change not to drift, but to true polar wander as proposed by Evans [38]. However, this requires

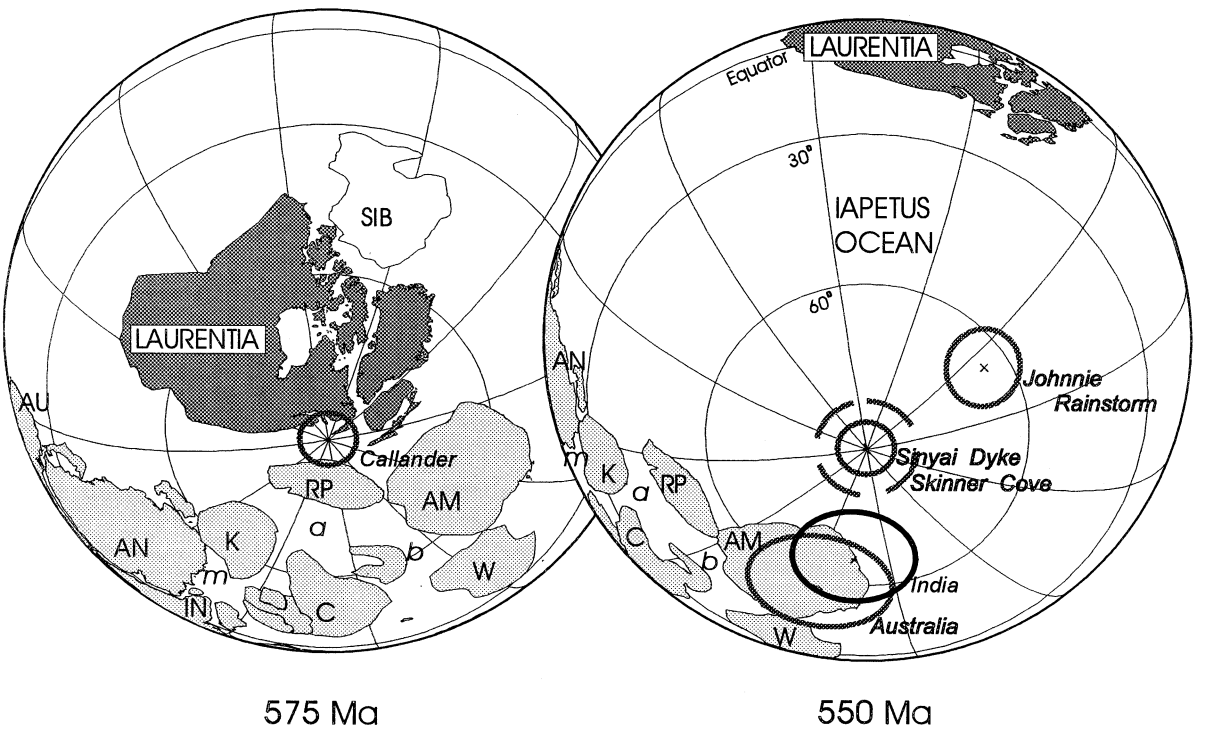


Fig. 9. Late Neoproterozoic paleogeography for ∼575 Ma and ∼550 Ma, modified from Dalziel, [2]. Poles are plotted with circles of 95% confidence. Abbreviations: *a* = Adamaster Ocean; *b* = Brazilide Ocean/Orogen; *m* = Mozambique Ocean/Orogen; *AM* = Amazonia; *AN* = Antarctica; *AU* = Australia; *C* = Congo–Sao Francisco; *IN* = India; *K* = Kalahari; *RP* = Rio de la Plata; *SIB* = Siberia; *W* = West Africa. Baltica is omitted for clarity. Diagram created with the assistance of the University of Texas Institute for Geophysics PLATES software.

Gondwana to share Laurentia's rapid paleolatitude change, conflicting with paleomagnetic data suggesting Gondwana remained relatively stationary [39]. The proposal [39] of ∼90º of true polar wander in the Early to Middle Cambrian is not supported by the Skinner Cove paleolatitude. We shall maintain the usual assumption of paleomagnetists that true polar wander is small (amounting to only ∼20º in the last 200 million years [37]) unless otherwise proven.

7. Implications for timing of Iapetus opening

Paleomagnetism (Fig. 8) suggests that between ∼570 Ma and ∼550 Ma Laurentia moved very rapidly northward to equatorial paleolatitudes where it remained through the Cambrian. We suggest that the beginning of this rapid northward drift of Laurentia at ∼570 Ma may represent the beginning of Iapetus sea-floor spreading between Laurentia and Gondwana. Laurentia's conjugate margin is uncertain but was likely the proto-Andean margin of Gondwana along the Rio de la Plata–Amazonia cratons [2].

Gondwanan paleomagnetism supports the hypothesis that Iapetus sea-floor spreading began at ∼570 Ma rather than at ∼550 Ma. Paleopoles from the Mozambique belt (Congo), India and Australia (summarized in [40]) place the proto-Andean margin of Gondwana near the south pole at ∼550 Ma when its Laurentian conjugate margin was ∼20ºS of the equator (Fig. 9). This assumes that no large separations remained between these Gondwanan cratons at 550 Ma. Other Gondwanan poles add support by suggesting that Amazonia moved across the south pole during the Cambrian [40].

Is this hypothesis of rapid opening of the Iapetus between ∼570 and 550 Ma consistent with

the stratigraphic and magmatic record along Laurentia's Iapetus margin? Evidence of rifting that was likely related to incipient opening of the Iapetus includes tholeiitic dyke swarms and continental flood basalts emplaced between 615 and 570 Ma and rift grabens with plume-related 580–560 Ma syenitic–carbonatitic alkali intrusions [23]. However, initiation of Iapetus sea-floor spreading is difficult to date precisely.

Subsidence curve analysis [1] of early Paleozoic Laurentian margin sedimentation indicates that thermal subsidence began between ~600 Ma and ~550 Ma. Alkali magmatism at ~555–550 Ma along the Laurentian margin was followed by transgressive sedimentation at the base of the Cambrian that is interpreted [41] as marking the rift to drift transition at ~545 Ma (timescale of Ref. [30]).

However, for two reasons, transgression at ~545 Ma gives only a minimum age for the start of drift. First, this transgression may only represent the shallow part of the basin above or cratonward of the hinge zone (as in western Newfoundland, where the overlying Cambrian to mid-Ordovician stratigraphic section is only ~2 km thick [41]). Here, transgression may have occurred as much as 40 Myr later than in the deep basin seaward of the hinge zone, judging by transgression above the hinge zone of North America's Atlantic margin that occurred 20 to 40 Myr after Atlantic sea-floor spreading began [42]. Second, even seaward of the hinge zone, theory suggests that transgression can be delayed by as much as 30 Myr if the margin is buoyantly underplated during rifting, by magmatism from an anomalously hot asthenosphere such as near a mantle plume [43]. Evidence for underplating of the margin when the Iapetus opened is provided by the low velocity layer in the lower crust beneath western Newfoundland [44].

Stratigraphic and subsidence curve analyses of Laurentia's Iapetan margin thus do not rule out the rift to drift transition being as old as the ~570 Ma Catoctin flood basalts. Younger, more alkalic magmatism represented by the ~564 Ma Mt. Rigaud syenite [45] and Sept Iles anorthosite [36], may be related to failed Ottawa–Bonnechere rift arm activity. The ~555–550 Ma Skinner Cove volcanics and Tibbit Hill metavolcanics likely represent Iapetan passive margin alkalic magmatism, analogous to Atlantic passive margin alkalic magmatism during the early Cretaceous [24,46].

8. Late Neoproterozoic paleogeography

Fig. 9 shows our suggested paleogeographic reconstructions for ~575 Ma and for ~550 Ma. These seek to reconcile latest Neoproterozoic paleomagnetic and geologic data from constituent cratons [2,5,40,47] assuming that the Skinner Cove Formation originated near Laurentia's Iapetus margin and that there was no large true polar wander.

For the ~575 Ma reconstruction, Laurentia is shown rifting from Amazonia–Rio de la Plata near the south pole. Paleomagnetic control is provided by the 577 Ma Callander Complex pole of Laurentia [33]. Relations between cratons follow those of Dalziel [2], although there are many alternative interpretations (e.g., [4,40]).

The ~550 Ma reconstruction resembles that of Torsvik et al. [5]. The Iapetus Ocean had opened between Laurentia, West Gondwana, and Baltica (not in view). Laurentian margin ~555–550 Ma alkalic magmatism (Tibbit Hill metavolcanics, Skinner Cove volcanics) postdated rifting, perhaps erupting in response to major rearrangement of plate boundaries and motions (e.g., [46]), marked by cessation of Laurentia's rapid northward drift; Laurentia may have stalled at the equator in a geoid low over a mantle sink [8]. Gondwana is shown consolidated, although its final consolidation may not have been until ~530 Ma [40]. The depicted separation across the Iapetus Ocean is based on the Skinner Cove Formation pole, and on the Sinyai Dyke pole of the Mozambique belt [48], which position Laurentia and the proto-Andean margin of Gondwana, respectively. Other ~550 Ma poles for Laurentia (Johnnie Rainstorm) and Gondwana (mean poles for India and for Australia) allow for a lesser but still wide (~5000 km) Iapetus Ocean (e.g., [5]).

Our reconstructions allow for the possible fleeting existence of a Pannotia supercontinent between the unifying ~630 Ma collision of Amazonia with Congo–Sao Francisco cratons [47] and the rifting of Baltica and Laurentia–Siberia from the consolidating Gondwanan cratons between ~600 Ma and ~565 Ma [5]. The existence of Pannotia also de-

pends on the timing of Gondwana assembly, which had taken place by ~530 Ma [40] but may have been earlier [48]. However, association of Laurentia (or Baltica [49]) with an assembled Gondwana in a Pannotia supercontinent is not paleomagnetically viable by ~550 Ma.

9. Conclusion

Flows and dykes of the 550 Ma Skinner Cove Formation in western Newfoundland carry a characteristic 'A' remanence that is shown to be primary by an intraformational conglomerate test. The paleolatitude calculated from the ten tilt-corrected 'A' site virtual geomagnetic poles is 19°S ± 9°. This is interpreted as the paleolatitude of the Iapetan margin of Laurentia at ~550 Ma, based on geological and geochemical evidence that the Skinner Cove Formation very likely formed close to this margin.

Comparison with other paleomagnetic results from Laurentia (Fig. 8) suggests that its Iapetan margin drifted rapidly northward from near the south pole beginning at ~570 Ma and reached ~20°S by 550 Ma where it remained through the Cambro-Ordovician. We suggest that this rapid northward drift of Laurentia opened a wide Iapetus Ocean between ~570 and 550 Ma (Fig. 9).

Laurentia's very high drift rate between ~570 and 550 Ma, implied by paleomagnetism (Fig. 8), requires testing with further paleomagnetic data from Laurentia (particularly for ~550 Ma, because it is not certain that the Skinner Cove Formation represents Laurentia). Paleomagnetic data from the Amazonia and Rio de la Plata cratons are also needed to test their role as conjugate margins to Laurentia in the formation of the Iapetus. This may require a resolution of magnetization age to within a few million years (as provided by the Skinner Cove volcanics and the Callander Complex) because the late Neoproterozoic was a period of rapid cratonic movement (and possibly of rapid true polar wander).

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